CARLOS DANIEL CEBALLOS AMOR

Full Stack Web Developer | Frontend · Backend · Mobile

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PROFESSIONAL SUMMARY

Full Stack Web Developer with 3+ years of end-to-end experience across Frontend (React, Next.js, Astro, Tailwind CSS), Backend (Python, Django REST Framework, NestJS, TypeScript) and Mobile (Flutter, Android TV). Strong background in designing, modeling and optimizing SQL/NoSQL databases (PostgreSQL, MySQL, MongoDB), process automation (ETL and RPA) and integrating AI agents (MCP protocol) for accounting, sales, customer service and analytics workflows. Proven track record scaling a B2B booking platform used by 1,000+ travel agencies and 1,400+ active users, building multi-tenant SaaS architecture with PostgreSQL Row-Level Security, and shipping e-commerce and internal business systems with payment-gateway and electronic-invoicing integrations. Focused on measurable results, data automation, test-driven quality and building scalable digital products.

PROFESSIONAL EXPERIENCE

Full Stack Developer (Freelance) — CATOCA **2026 – Present**

CATOCA — Colombian coffee export company · catoca.cafe

Brand Website & Online Store

- Built the company's brand website and online store (live at catoca.cafe): brand storytelling and origin pages, product catalog, cart, checkout and contact form, using React 19, Vite, TypeScript, Tailwind CSS v4 and anime.js animations.
- Developed an Express backend for checkout and email, integrating the Colombian payment gateway **Wompi** with server-side payment confirmation instead of trusting the client.

Internal Sales, Inventory & Commissions System

- Designed and built the company's internal platform to record sales (web store and sales reps), control inventory, calculate commissions and generate DIAN-compliant pre-invoices, as a modular monolith with NestJS + Fastify, Drizzle ORM and PostgreSQL 18, and a React 19 + TanStack Router/Query frontend sharing Zod contracts.
- Modeled lot-based inventory (lot and roast dates) with FIFO stock output and a movement-based kardex; commissions with rate snapshots and approval/payout workflow; versioned VAT configuration; immutable sales with admin-only, audited revisions (before/after) and role-based access.
- Delivered consolidated Excel accounting exports for **Siigo Nube** and integrated the online store through an orders API with API keys and a Wompi webhook, so web sales automatically update inventory, pre-invoices and reports.
- Enforced 100% test coverage (unit, component and API tests against real PostgreSQL) plus Playwright E2E tests; production deployment with Docker Compose and Nginx (TLS, rate limiting).

Founder & Lead Developer — Pargo (SaaS) **July 2026 – Present**

Personal project · Multi-tenant POS SaaS for restaurants and hotels on Colombia's Caribbean coast · built with AI-assisted development

- Architected and built a multi-tenant SaaS platform (POS with offline mode, DIAN electronic invoicing, digital menus, web presence and AI-driven reputation management) as a TypeScript monorepo: NestJS modular monolith, PostgreSQL with **Row-Level Security** tenant isolation, an isolated public API with its own database role, and a background worker using a transactional outbox (pg-boss).
- Delivered the POS core: floor plan by zones and tables, orders and kitchen tickets, real-time Kitchen Display System over WebSockets (Valkey/Redis pub/sub), transactional sales and payments, cash-register opening, blind count and Z report, refunds, voids, supervisor-approved discounts and bill splitting.
- Implemented an offline-capable POS (PWA with signed sync queue), electronic invoicing pipeline (Factus/DIAN), local receipt printing, and recipe-based inventory with per-location ledger, weighted average cost, purchases, waste and physical counts.
- Built the backoffice (React 19, Vite, TanStack, Tailwind) and public digital-menu website (Next.js with SSR/ISR for SEO); security by design with Argon2id, TOTP MFA, RBAC, device keys and operator PINs for POS terminals, OWASP-based threat model, OpenTelemetry and Trivy-scanned Docker images.

- Built with AI-assisted development as a single developer: AI speeds up implementation, while the architecture and every technical decision are mine and justified in ADRs — a modular monolith that one person can deploy and operate, database-enforced tenant isolation with RLS instead of per-query filters, an offline POS because internet outages must not stop sales, and an auditable inventory ledger.
- Maintained 686 automated tests (Vitest + Testcontainers on real PostgreSQL) plus Playwright E2E, with architecture decisions documented as ADRs; MVP ready for client demos.

Full Stack Web Developer **February 2025 – June 2026**

GEH Suites — Cartagena, Colombia

Backend & Databases

- Scaled, maintained and extended the API of **Booking Connect**, the company's B2B hotel availability and booking platform (NestJS + **MongoDB**), used by 1,000+ travel agencies and 1,400+ active users, including a complete flight-booking API.
- Designed, modeled and optimized **PostgreSQL** relational databases (schemas, advanced SQL queries and indexing) for the company's internal systems: multi-hotel expense management and hotel occupancy ETL.
- Developed robust, scalable RESTful APIs with Django REST Framework and NestJS, improving integration between internal and external systems.
- Built daily ETL processes in Python and PostgreSQL to process hotel occupancy data from multiple sources, with real-time statistical reporting.
- Designed and implemented a multi-hotel financial expense management system (13 properties) with expense tracking, petty-cash management, real-time statistics and automated PDF and Excel reports, backed by PostgreSQL (Django REST Framework).
- Automated administrative invoicing processes with an RPA bot (invoice extraction from email, CRM registration and attachment upload), reducing operating time by 40%.
- Refactored legacy code and optimized backend processes, improving overall platform performance and scalability.

Frontend

- Developed modern, responsive, high-performance interfaces with React, Astro and Tailwind CSS for Booking Connect and the internal systems, focused on usability for 1,400+ active users.

Mobile Development

- Designed and developed a cross-platform (iOS/Android) Flutter app for travel agencies that operate mainly from mobile devices, streamlining their on-the-go booking flows.
- Developed an Android TV app with Flutter integrating virtual intercom, room service and IPTV managed from the TV through RESTful APIs.

Artificial Intelligence & Automation

- Designed and deployed multiple AI agents (MCP protocol and OpenCLAW) to automate accounting, sales, customer service, HR and statistical analysis processes for the B2B platform (availability, bookings and quotes).

Full Stack Web Developer **January 2023 – January 2025**

Caritech — software development for the company's clients

- Built full stack web applications for the company's clients end to end, from requirements and data model to deployment and maintenance.
- Designed, modeled and optimized relational and NoSQL databases (PostgreSQL, MySQL, MongoDB) per client: schemas, queries and indexing.
- Developed RESTful APIs with Python (Django REST Framework) and Node.js to integrate client systems with internal and third-party services.
- Built responsive interfaces with React, JavaScript, HTML and CSS for client web platforms and administrative panels.
- Automated data processes and reporting with Python scripts (ETL, PDF and Excel reports) for client operations.
- Developed cross-platform mobile apps with Flutter for clients that needed a mobile channel.

Flutter Development Lead — Caminante Project **January 2023 – December 2023**

Independent

- Led the development of a Flutter mobile app focused on tourism in Cartagena de Indias, coordinating the technical work from start to finish.
- Designed a tourist-centered user experience (UX) with features adapted to the local environment and context.

- Recognized as a **high-impact project by SENA's Fondo Emprender**, and took part in a **public tender held by the city of Cartagena**.

TECHNICAL SKILLS

- **Frontend:** React, Next.js, Vue.js, Astro, Vite, TanStack Router/Query, Tailwind CSS, HTML, CSS, JavaScript, TypeScript.
- **Backend:** Python, Django, Django REST Framework, NestJS, Fastify, Express, Node.js, RESTful API design, WebSockets, Zod.
- **Mobile:** Flutter, Dart, cross-platform development (iOS/Android), Android TV integration, PWA.
- **Databases:** SQL, PostgreSQL (Row-Level Security, multi-tenancy), MySQL, MongoDB, Redis/Valkey, Drizzle ORM, relational modeling, query optimization, indexing, ETL processes.
- **Integrations:** Payment gateways (Wompi), DIAN electronic invoicing, Siigo Nube accounting exports, CRM (Bitrix24), webhooks and API keys.
- **Artificial Intelligence:** AI agent development, MCP protocol, OpenCLAW, AI-driven automation, AI-assisted development, Machine Learning (fundamentals).
- **DevOps, Testing & Cloud:** Git, GitHub, Docker, Docker Compose, Nginx, AWS, OpenTelemetry, Vitest, Testcontainers, Playwright, RPA, PDF/Excel reporting, Agile methodologies.

EDUCATION & TRAINING

- Systems Engineering (in progress) — Universidad de Cartagena.
- Business Strategy Management in the Digital Era — Universidad Tecnológica de Bolívar, 2021.
- Full Stack Development Course — edX, 2023.
- Introduction to Machine Learning — Udemy, 2024.

LANGUAGES

Spanish: Native. English: B2 (upper-intermediate).

SOFT SKILLS

Technical leadership, effective communication, decision-making, teamwork, adaptability and results orientation.

REFERENCES

Professional references from GEH Suites and Caritech available upon request.